

Systems for Machine Learning (Stanford CS229S)

Stanford Frontier AI Learning System

Baseline: October 6, 2026. Print companion to the HTML course site:
crash course and cheatsheet.

Contents

Crash course: the whole course in thirty minutes	2
U01: the transformer workload	2
U02: hardware-aware design	2
U03: transformer accounting and speculative decoding	2
U04: CUDA and efficient attention	2
U05: quantization, sparsity, structured operators	2
U06: training, adaptation, data pipelines	3
U07: linear attention, SSMs, FFT	3
U08: serving and sparse MoE	3
U09: parallelism, clusters, scheduling	3
U10: retrieval, guests, synthesis	3
The one-page decision list	4
Cheatsheet: formulas, shapes, traps, decision rules	5
Shapes	5
Cost formulas	5
Quantization	5
Adaptation	6
Serving	6
Retrieval	6
Scheduling	6
Evidence grades	6
Defense five	6

Crash course: the whole course in thirty minutes

Date: 2026-10-06. Baseline: October 6, 2026. Condensed from all 10 units (U01-U10). Every number is a lesson toy computed in the unit lessons, not a vendor claim. Claim class: PLANNED / SOURCE ATTRIBUTION PENDING unless a calendar anchor is named. Calendar anchors: Week 1 Sep 27 (U01), Week 2 Sep 30 / Oct 04 (U02, U03), Week 3 Oct 07 / Oct 11 (U04), Week 4 Oct 14 (U05), Oct 18 / 21 / 25 (U06), Week 6 Oct 28 / Nov 01 (U07), Week 7 Nov 04 / Nov 08 (U08), Oct 25 / Oct 30 / Nov 15 / Nov 18 (U09), Week 10 Dec 02 (U10).

U01: the transformer workload

A transformer block is attention plus MLP. Per token per layer: attention $\sim 4n^2 + 4Tn$ (projections plus scores), MLP $\sim 8n^2$ (up+down with $d_{ff}=4n$). KV cache: $2*L*T*n$ bytes per float. Decode is memory-bound. Prefill is compute-bound. Rule: count FLOPs and bytes before touching the machine.

U02: hardware-aware design

GPUs are a memory hierarchy: registers, shared memory, HBM. Arithmetic intensity $I = \text{FLOPs/byte}$ decides the bound. The roofline caps attainable FLOP/s at $\min(\text{peak}, I * \text{bandwidth})$. Kernels fuse to keep tensors on chip. Launch costs and occupancy decide whether the fusion pays. Rule: measure intensity first, optimize second.

U03: transformer accounting and speculative decoding

Training: $6*N*D$ FLOPs. Decode: $2*N$ per token plus KV traffic. KV bytes: $2*L*T*n$ per float. Batching amortizes weight reads: throughput rises sublinearly. Speculative decoding: a draft model proposes, the target verifies. Exactness holds because verification samples from the target. Acceptance rate sets the speedup. Rule: decode is a memory problem. Speculation trades draft compute for fewer target passes.

U04: CUDA and efficient attention

Threads form warps (32), warps form blocks. Coalesced access merges transactions. Tiling fits blocks in shared memory. Online softmax streams with a running max. FlashAttention tiles attention so the $T \times T$ matrix never materializes: exact, IO-aware. Backward recomputes instead of storing. Rule: never materialize what you can stream.

U05: quantization, sparsity, structured operators

Linear quantization: $s = (\text{max}-\text{min})/(2^b-1)$, $z = \text{round}(-\text{min}/s)$. In-range error $\leq s/2$. Per-channel beats per-tensor on skewed weights (218x on the quiet channel in the toy). 2:4 sparsity is the pattern hardware likes. Butterfly: $2*n*\log_2(n)$ params with full mixing. Monarch generalizes it. Ship gates: perplexity, task, latency. Rule: shrink the bytes, prove the quality.

U06: training, adaptation, data pipelines

Loss is linear in log-log space: fit two points, predict the third (slope -0.0748/decade in the toy). Few-shot conditions without weight updates. threshold metrics manufacture emergence jumps. SFT masks the instruction. RLHF climbs reward with a KL leash. Constitutions move human input to the rules. LoRA trains $r(d+k)$ params and merges free. The pipeline runs at the slowest stage. Packing lifts real-token throughput 1.56x in the toy. report mean plus seed spread. Rule: predict in log space, gate on quality, seed everything.

U07: linear attention, SSMs, FFT

Quadratic attention: $2T^2d$ FLOPs, T^2 score memory (256 MiB fp32 per head at $T=8192$). Linear attention regroups: $\phi(Q)(\phi(K)^T V)$, $O(Td^2)$, 64x fewer FLOPs in the toy. SSM: $h = Ah + Bx$, $y = Ch$. Unrolls to a convolution kernel $K_k = CA^kB$. Scans parallelize it. FFT: conv in $O(n \log n)$, 33x at $n=1024$. Stability: A^n compounds. fp16 dies at step 1115 for $A=1.01$. Price: d^2 state cannot hold T exact slots (64x short in the toy). Rule: pay T^2 for exact routing or Td^2 for a blur. Grade guest claims by artifact.

U08: serving and sparse MoE

Continuous batching boards on arrival (1.0 vs 1.5 s mean latency in the toy). Prefill is a gulp, decode is sips: chunk the gulp. KV: 0.5 MiB per token in the toy ($L=32$, $n=4096$, fp16). Report p99, not means. MoE: top-k routing, capacity = tokens*k/experts*factor, drops past capacity, imbalance = max/mean (5x in the toy). Dispatch: tokens*k*d bytes each way. Batching bends toward an asymptote. Cost = price/rate. Rule: every win lands in tokens per second.

U09: parallelism, clusters, scheduling

Four splits: data, tensor, pipeline, expert. Each has a tax (sync, collectives, bubbles, dispatch). Ring all-reduce: $2(n-1)/n \cdot S/BW$ (0.35 s in the toy). Keep TP inside the node (12x wire gap in the toy). Bubbles: $(p-1)/(m+p-1)$. ZeRO-3: 16 bytes* P/n per GPU. Checkpoint: $(N/k) \cdot \text{cost}$. Straggler tax: $(\text{max}-\text{median})/\text{median}$. DRF: max demand ratio. MFU: achieved/peak. Cluster MTBF: per-GPU/n (4.3 days at 256 in the toy). Rule: name every tax with a formula.

U10: retrieval, guests, synthesis

Index: amortize build over queries (break-even 160k in the toy). Dense 2.86 GiB vs PQ 61 MiB (48x). ANN: set the recall floor, then pick. RRF fuses by rank. Cascade: 405 ms, mostly rerank. Grade guest claims: measured > derived > stated > opinion. Proposals need failure criteria. Profile from a hypothesis. Seeds: mean plus spread. Gate: $\delta \geq -\epsilon$. Defend without bluffing. Rule: build it, measure it, gate it, defend it.

The one-page decision list

1. Count FLOPs and bytes first (U01-U03).
2. Never materialize T^2 (U04, U07).
3. Shrink bytes, prove quality (U05).
4. Fit laws, mask SFT, leash RL, seed runs (U06).
5. Batch continuously, chunk prefills, watch p99 (U08).
6. Split wisely, mind the thin link (U09).
7. Amortize indexes, fuse ranks, gate quality (U10).
8. Grade every claim by its artifact (all units).

Cheatsheet: formulas, shapes, traps, decision rules

Date: 2026-10-06. All 10 units. Toy values from the lessons. "toy" marks computed examples, not specs.

Shapes

Symbol	Shape
X	(B, T, n)
Q, K, V	(B, h, T, d)
W _{q,k,v,o}	(n, n)
W ₁ / W ₂	(n, d _{ff}) / (d _{ff} , n)
scores	(B, h, T, T)

Cost formulas

What	Formula	Toy
Train FLOPs	6 N D	1.2e20 (1B, 20B tok)
Decode/token	2 N + KV traffic	-
Attention scores	2 T ² d / head	17.2 TFLOPs (8k, 128)
Score memory	T ² floats	256 MiB fp32 (8k)
KV/token	2 L n bytes/float	0.5 MiB (32, 4096, fp16)
KV total	2 L T n bytes/float	-
Linear attn	2 T d ²	0.268 GFLOPs (64x less)
SSM state	d ² floats	64 KiB
FFT conv	3 n log ₂ n	33x at n=1024
Ring all-reduce	2(n-1)/n S/BW	0.35 s (20 GB, 8, 100 GB/s)
Bubble	(p-1)/(m+p-1)	27.3% (4, 8)
ZeRO-3/GPU	16 P / n bytes	20.0 GB (10B, 8)
Checkpoint	(N/k) c	300 s (1000, 100, 30 s)
MTBF cluster	per-GPU / n	4.3 days (256)
MFU	achieved / peak	48.1% (150/312)

Quantization

$s = (\max - \min) / (2^b - 1)$. $Z = \text{round}(-\min/s)$. $x_q = \text{clamp}(\text{round}(x/s) + z, 0, 2^b - 1)$. $\hat{x} = s(x_q - z)$. In-range error $\leq s/2$.

Adaptation

Scaling: $\log_{10} L = a + b \log_{10} N$ (toy $b = -0.0748/\text{decade}$). SFT: loss on response tokens only.
RLHF: $E[r] - \beta \text{KL}$. LoRA: $W_0 + BA, r(d+k)$ params (128x fewer at 4096/16).

Serving

Step = max(compute, fetch+assemble). P99 over means. Capacity = tokens*k/experts*factor.
Dispatch = tokens*k*d bytes one way. $R(B) = B/(c_0 + c_1 B)$. $\$/\text{MTok} = 1e6/R * \$/\text{hr} / 3600$.

Retrieval

Amortized = $B/N + q$. RRF = $\sum 1/(k + \text{rank})$. Cascade = $\sum \text{count} * \text{cost}$. Gate: $\delta \geq -\epsilon$.

Scheduling

DRF share = $\max(\text{demand}/\text{total})$. Straggler tax = $(\text{max} - \text{median})/\text{median}$. Makespan via FIFO sim.

Evidence grades

measured > derived > stated > opinion. Never promote without the artifact.

Defense five

claim, evidence, baseline, failure, cost.